Process Technology Enablers for Aggressive Scaling of Hybrid Bonding Temperature and Pitch

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***Abstract*— Modern applications, such as artificial intelligence (AI), 5G communications, and autonomous vehicles, all require high-performance chips to supply enormous computing power and memory capacity. But it is becoming extremely difficult and prohibitively expensive to keep shrinking transistors into angstrom-scale. Therefore, the semiconductor industry is eyeing advanced packaging as an alternative way to continue the roadmap of chip performance advancement. Hybrid bonding is an essential technology in advanced packaging due to its multiple advantages in enhancing system level performance; for instance, enabling 3D stacking and heterogenous integration, higher I/O density, faster speed, lower power consumption, and better thermal management, etc. Hybrid bonding is often done at 350-400°C, to ensure sufficient Cu expansion and diffusion between adjacent wafers or chips to achieve seamless bonding. However, for some applications such as SoIC and advanced generation high-bandwidth memory (HBM), it is necessary to lower the bonding and post-bonding processing temperatures well below 350°C because temporary bonding adhesive materials are used in the integration flows, and these adhesives typically have 200-300°C as their upper working temperature limit. Low temperature may also be required to avoid damaging the functionality or reliability of underlying devices. We have previously reported a high electrical yield robust integration test vehicle for wafer-to-wafer hybrid bonding with SiCN as the bonding film at various tight bond pad pitches (0.5 – 1.2 µm) [1, 2]. In this work, we attempted to use this W2W hybrid bonding test vehicle to evaluate two process technology solutions for enabling the scaling of hybrid bonding temperatures down to 200 °C regime, at deeply scaled pitches. One method is to carefully optimize the SiCN bonding film properties and meticulously control the bonding surface topography by advanced CMP process control. Another approach evaluated is to boost the Cu diffusivity at low temperatures by applying a thin interfacial metal diffusion layer on top of Cu bonding pads with a proprietary electro-less selective deposition process [3]. Whole wafer electrical results of wafers that adopted these process techniques were analyzed. For the first time, we demonstrated electrical yield enhancement of 1 million-link via chains at 500 nm pitch by using an interfacial metal diffusion layer, with bonding and post-bonding maximal temperature as low as 200 °C. An interfacial metal technology is a promising approach to enable hybrid bonding manufacturability at low temperatures.**

Keywords*— *Hybrid bonding, wafer-to-wafer bonding, W2W, die-to-wafer bonding, D2W, post bonding anneal, low temperature bonding, tight pitch, interfacial metal diffusion layer, SiCN, CMP

I. Introduction

With Moore's law slowing down and technology node scaling approaching its physical limits, advanced packaging technology has become more important than ever to sustain the system level performance advancement that is necessary for the ever-increasing computing needs driven by computation-intensive applications such as AI, high-performance computing, robotics, and self-driving cars. Direct wafer-to-wafer, die-to-wafer, and chip-to-chip hybrid bonding have become key technologies for advanced semiconductor packaging. They enable heterogenous integration of chips or wafers with different functionalities and provide system performance benefits including wider bandwidth, higher array efficiency, higher I/O density, lower power consumption, smaller form factors, shorter fabrication cycle time, and overall higher performance. In Yole Group's opinion, when interconnect pitches shrink to less than 10 µm, hybrid bonding is going to be the only reliable technique for such interconnect density scale, comparing to micro-bumps [4]. And the interconnect pitches are expected to shrink much further into the sub-micron range, which has been actively studied by many research groups globally in recent years [1, 2, 5-7].

With AI making big strides every day and its demand for high performance computing growing rapidly, high bandwidth memory (HBM) has risen as a game changer to offer superior bandwidth and power efficiency benefits comparing to other traditional memories. Major HBM manufacturers have recently confirmed their plans to adopt the hybrid bonding technology in

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their HBM5 20-Hi stack generation products, says TrendForce [8]. For hybrid bonding to be applied in HBM, it is generally agreed that the highest processing temperature for the entire integration flow should be lower than the current typically used around 350 °C range. This includes post-bonding anneal and post-bonding integration processing steps. Lowering the bonding and post-bonding processing temperatures for HBM comes from the necessity to comply with the maximal temperature allowance of the temporary bonding adhesives, usually around 200 - 300 °C, and from the concerns of degrading sensitive underlying devices. The low temperature requirement applies to many other D2W and D2D bonding applications for similar reasons. Lower temperature also brings other benefits such as reduced thermal stress, etc.

We have previously reported a robust integration flow with high electrical yield for 300 mm wafer-to-wafer hybrid bonding with SiCN bonding interface at 0.5 µm pitch [1, 2]. High yield electrical results up to 1 M via chains were demonstrated by bonding the wafers at room temperature, followed by a 350 °C post-bonding anneal to allow for Cu expansion and diffusion to form strong Cu-Cu bonds. In this work, we evaluate two process approaches to lower the hybrid bonding integration thermal budget to 200 °C.

One approach is mainly by optimizing the bonding layer CMP process, to obtain desired bonding surface topographies for high quality bonding, without adopting novel Cu metallurgy techniques such as nano-twinned Cu, fine-grained Cu, and interfacial passivation layers [9-11]. CMP process is critically important for bonding, especially for sub-µm level tight pitches and when thermal budget is significantly lowered so less thermal energy is available for Cu to expand or diffuse.

Another method is to apply an interfacial metal diffusion layer on top of the Cu bonding pads. This interfacial metal layer is expected to lower the Cu diffusion activation energy and therefore would significantly boost the Cu diffusivity and the diffusion length through this layer, comparing to Cu self-diffusion through itself. This mechanism is believed to be able to lower the hybrid bonding temperature.

II. EXPERIMENTAL

A. Test vehicle

To study the aggressive scaling of hybrid bonding pitch and temperature, we used a previously reported integration test vehicle which demonstrated the capability of achieving very high whole wafer electrical yield for 300 mm wafer-to-wafer hybrid bonding with SiCN bonding film at 500 nm pitch [1, 2]. Highly yielding electrical results for up to 1M via chains were shown. This test vehicle has a built-in DOE in various combinations of device feature sizes in bonding pitch (0.5 - 1.2 µm), bonding via pad dimension (200 - 600 nm), Cu areal density (4 - 25%), and different via chain lengths from 1 Kelvin via to 1 million link daisy chains. The sketch (Fig. 1) of the test vehicle is reproduced [1] below for illustration to help describe the current work.

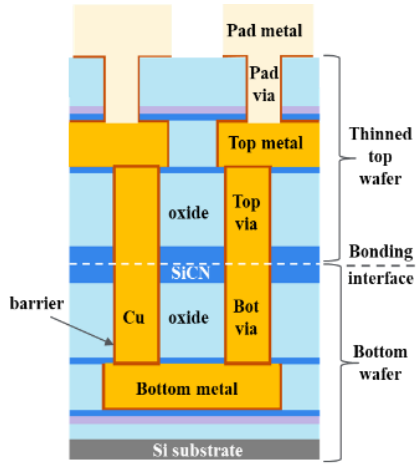

Fig. 1. Illustration of the 6-layer W2W hybrid bonding test vehicle used in this work. The bonding layer used a SiCN film with optimized composition for high bonding energy. The top and bottom via sizes may be the same or different.

B. SiCN bonding films

Selecting a strong and stable bonding film is one of the crucial prerequisites to warrant a successful bonding. When dealing with the deeply shrunk sub-µm scale bonding pitches, dielectric films with much stronger bonding energy than the oxides used in previous looser size applications are needed, to bear up against the higher stress exerted by the smaller dielectric spacings between the Cu bonding pads. When the Cu pads draw closer to each other, Cu diffusion concerns need to be taken into consideration too. After thoroughly evaluating different bonding dielectrics, composition-optimized SiCN films deposited at 200 °C to 350 °C temperature range were chosen as the bonding films for our study. The key benefits of using these SiCN films are their excellent Cu diffusion barrier property and high bonding strength [1, 2].

C. Bonding layer CMP

One can never overstate the importance of bonding layer planarization by chemical-mechanical polish (CMP) in engineering the bonding surface topography, which crucially determines the final hybrid bonding quality. To achieve strong and reliable dielectric bonding, a particle-free, smooth, and flat dielectric surface is needed in addition to high bonding energy and thermal stability requirements. For high whole wafer yield including the wafer edge area, CMP edge roll-off needs to be minimized too. For Cu pads between wafers or chips to bond seamlessly, the CMP process needs to be carefully controlled so it precisely renders small Cu pad dishing, with values appropriate to the pattern designs, uniformly across the whole wafer. This dishing control also needs to be repeatable from wafer to wafer. When the bonding pad pitches scale to the aggressively small sub-µm regime, and when the bonding temperature needs to be lowered, tight dishing and topography control becomes more important because Cu expansion and diffusion are limited due to less thermal energy supplied.

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With Applied Material's Reflexion® LK Prime CMP tool sets, there are many hardware, software, and process knobs to fine tune processes to meet the topography requirements for different purposes. For this work, we intentionally applied two different CMP processes to create different bonding layer surface finishes with reasons to be elaborated in the "results and discussion" session of this article. One is an optimized CMP process by adopting state-of-the-art advanced tool configurations with proper types of polishing pads and heads, multi-zone pressure control, and advanced in-situ metrology and process control algorithms. The other one is a less optimal legacy process by not utilizing the advanced process control features. Fig. 2. shows the different topographies achieved by using these two types of CMP processes [1]. For both processes, the planned ~ 1 nm Cu dishing could be generated. But the bonding dielectric (SiCN) terrain is different. One has a flat top and the other one is rounded. Process (a) also has excellent within wafer uniformity (< 1 nm dishing variation) and minimized edge roll-off. The larger dielectric roll-off slopes in process (b) could also deteriorate the bonding quality. The impact of the CMP process on the bonding layer topography was inevitably exhibited by the electrical yield performance which were analyzed in details in a previous report [1].

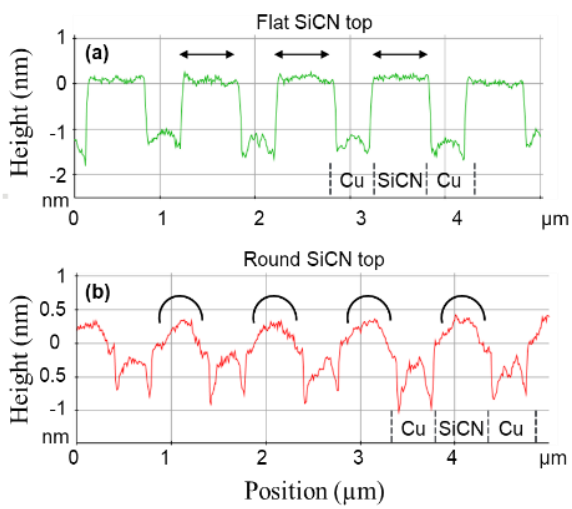

Fig. 2. Example AFM line scans of post-CMP Cu/SiCN bonding layer. (a) optimized CMP process by adopting hardware settings, state-of-the-art multi-zone control, and software and metrology knobs for topography control. Flat SiCN surface was achieved with minimized erosion. (b) non-optimal CMP process when hardware or process control parameters were not optimized. The SiCN top rounding and the larger dielectric roll-off slope are not desired for good bonding quality.

D. Electroless interfacial metal diffusion layer

Diffusion of copper atoms from one film into a copper layer of an adjacent film in intimate contact with the first film is advantageously driven by the application of both pressure, which helps to promote the intimate contact, and more importantly, temperature, which drives the diffusion rate of the Cu atoms. The diffusion rate of Cu atoms can also be enhanced by factors such as the Cu grains' crystalline orientations and the grain sizes. This triggered much study of altering the Cu metallurgy by using nano-twinned Cu with <111> orientation [9] and fine-grained Cu [10] to lower the hybrid bonding temperatures.

It is also known that Cu will diffuse more rapidly into some metals than it will when diffusing directly into another Cu film, because Cu diffusivity through these metals is much higher than its self-diffusion diffusivity due to the lower activation energy [11]. Such metals into which Cu will favorably diffuse include Ge, Pt, Au and Ag, etc. By creating a thin layer of such metals on top of either or both of the Cu layers to be bonded, the diffusion rate of Cu into this diffusion layer occurs more rapidly and occurs at a lower temperature than may be achieved if the two copper layers were brought into intimate contact with each other but absent the aforementioned diffusion layer. The key to the successful formation of the interfacial diffusion layer include the following criteria:

1. The diffusion layer must have higher diffusivity for the diffusion of Cu at a lower temperature and pressure than might be expected for a Cu to Cu contact alone.
2. The diffusion layer must be relatively thin so as to allow Cu to rapidly transit through it and pass to the adjoining Cu layer for bond formation.
3. The diffusion layer must be relatively smooth. The underlying Cu film must be very smooth to promote the atomically intimate interface between the adjoining Cu films. The introduction of the diffusion layer must maintain this intimate interface, so a lack of smoothness, pitting, hillocks, or non-uniformity must be avoided. This is best accomplished if the diffusion layer is thin in order to retain and propagate the Cu film roughness (or smoothness) into the interfacial layer.

Techniques such as atomic layer deposition (ALD) and physical vapor deposition (PVD) can create such interfacial films. But cost and throughput render such techniques to be less desirable for high volume manufacturing (HVM). We developed a proprietary electroless deposition process [3] as a more cost effective alternative to deposit the interfacial metal layers (abbreviated as "IF metal" in figures below) on top of Cu bonding pads.

The electroless metal deposition is an oxidation reduction (ReDOx) reaction whereby atoms of one metal (M) are reduced in a chemical solution to deposit on another metal (Cu).

As an generalized illustration of such a chemical reaction:

$$2M^{+1} + 2e^- \rightarrow 2M^0$$
$$Cu^0 \rightarrow Cu^{+2} + 2e^-$$

Although the above reactions will occur spontaneously when a copper film is introduced into a chemical solution containing the IF metal ions, the deposit is generally only loosely adhered to the underlying Cu and is of a poor quality and unacceptable for use in bond formation. The use of a certain class of additives to the solution containing the M^{+1} metal ions will enable the creation of a smoother and consistent layer which is suitable as a diffusion layer for the creation of Cu-Cu bonds at lower temperature than might otherwise be achieved.

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The smooth deposit is extremely favorable for hybrid bonding as it becomes self-limiting in nature. Once the Cu film is "capped" with a continuous layer of the IF metal, further deposition ceases as there is no further contact between metal ions and copper. We have found this self-limiting thickness to be in the range of ~ 2 – 4 nm, which appears to be well suited to the desired bonding application. Because the mechanism for the electroless deposition is one of an "exchange," the Cu dishing profile established by the CMP process is minimally altered, which is truly a blessing so we can avoid the otherwise necessary and potentially complex co-optimization between Cu dishing adjustment and metal deposition thickness.

Fig. 3 shows the TEM and EELS images of such an electroless deposition of an IF metal film onto the Cu pads of a hybrid bonding wafer, after the bonding layer CMP already completed. A continuous and uniform layer of the IF metal with approximately 2.5 nm thickness was selectively deposited only onto the places where Cu exists but not on top of the SiCN film. This fortunate selective deposition process with the Cu-replacement nature made the adoption of the IF metal layer into the hybrid bonding flow almost effortless. There is no need to remove the IF metal layer on top of the SiCN dielectric if the deposition was otherwise not selective, like in a PVD scenario. There is also no need to change the CMP process due to the addition of such a layer after the Cu CMP is already carefully developed with desired Cu dishing.

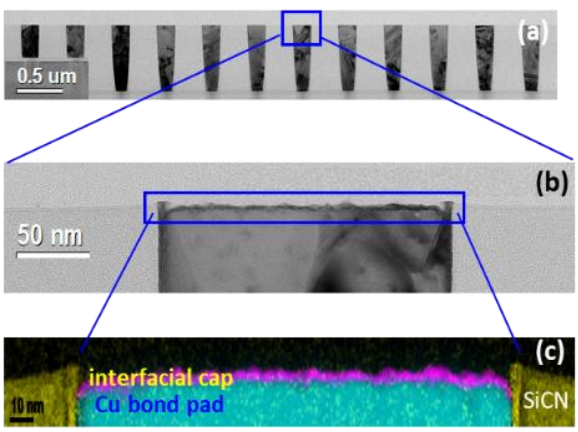

Fig. 3. TEM (a, b) and EELS (c) images showing a continuous and uniform, ~ 2.5 nm thick interfacial metal diffusion layer selectively deposited only on top of the Cu pads but not on SiCN.

E. Hybrid bonding and post-bonding integration

Wafers were bonded on an EVG Gemini FB XT platform with a SmartView NT3 alignment module. The pre-bonding wafers went through a plasma activation process and then a DI water clean, creating particle-free and surface hydrated finishes. The wafers were then aligned with each other through the alignment module and bonded at room temperature. The bonded wafers were then annealed at different temperatures, between 200 °C and 350 °C for this study, for the dielectrics to form strong covalent bonds and for Cu pads from the two adjacent wafers to connect.

Two points need to be made clear before we delve into the experiments and results discussion.

1. Wafer bonding is always conducted at room temperature first, followed by anneals at elevated temperatures. For simplicity, in following discussions we often call a bonding process as "200 °C bonding" or "280 °C bonding". This just meant that the post-bonding anneal temperatures were 200 °C or 280 °C.

2. For a "200 °C bonding", we carefully controlled all process steps in our test vehicle integration flow to ensure all post-bonding processing temperatures are equal or lower than 200 °C. Similarly for other temperatures.

III. RESULTS AND DISCUSSION

A. Low temperature bonding at 200 °C

We previously published [2] a group of TEM pictures for successful W2W SiCN/Cu bonding at different pitches (500 nm to 1.2 µm), with 350 °C post-bonding anneal temperature. Excellent bonding quality was demonstrated for a wide window of feature size DOE. In this work, we lowered the post-bonding

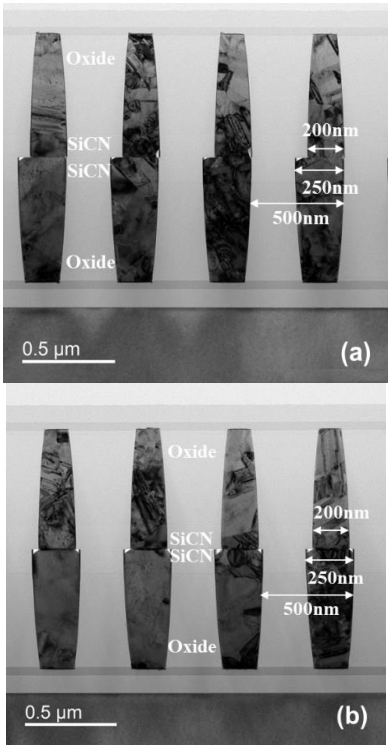

Fig. 4. TEM images of two 500 nm pitch W2W SiCN/Cu bonded via chains. In both bonded wafer pairs, the top wafer via CD was 200 nm and the bottom wafer via CD was 250 nm. Both wafers were bonded at room temperature and then annealed to 200 °C for 2 hours. Wafers were cut right after the post-bonding anneals. No further thermal process was done after the anneals. (a) No IF metal film was used. (b) a ~ 2.5 nm thick IF metal layer was electrolessly deposited on top of the Cu pads before bonding.

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anneal temperature down to 200 °C to observe the impact of the reduced temperature on bonding quality. With our two process approaches for bonding layer topography control and Cu diffusivity boost, namely CMP process optimization and application of an interfacial metal diffusion layer, we achieved similar high bonding quality at 200°C. Fig. 4 shows TEM pictures of 500 nm pitch via chains for two wafers annealed at 200 °C. Fig. 4 (a) was a wafer that directly bonded with typical Cu pads. Fig. 4 (b) was a wafer that a thin (~ 2.5 nm) interfacial metal diffusion layer was deposited on top of the Cu pads by using our electroless selective deposition technique. Void-free bonding was observed for both wafers.

TEM characterization demonstrated the possibility of maintaining bonding integrity when the bonding temperature scales down to 200 °C. But the examined area and the sampling rate of TEM are limited. Electrical testing was done to evaluate the whole wafer bonding quality.

B. Impact of CMP control on bonding temperature scaling

In a previous study [1], we demonstrated in detail how an optimized CMP process significantly improved the whole wafer electrical yield for all pitches included in the test vehicle design (0.5 - 1.2 µm range), at 350 °C. In this work, we applied both the optimized and the less-optimal CMP processes to the Cu/SiCN hybrid bonding wafers, and bonded them with 200 - 350 °C temperature splits. Wafers were fabricated by following the integration flow illustrated in Fig.1. The upper temperature limits for post-bonding integration steps were capped at the bonding temperature for each wafer.

Fig. 5 shows the whole wafer via chain resistance distribution comparison of daisy chains with 10 K, 100 K, and 1 million links for three wafers representative of two CMP processes and two bonding temperatures. The red curve was for a wafer polished by the optimized CMP process and annealed at 350 °C. The green curve was for a wafer polished with a fair legacy CMP process and annealed at 350 °C too. The blue curve was for a wafer polished with the optimized CMP process and annealed at 200 °C. The electrical yield differences demonstrated the overwhelmingly important role of CMP topography control in final performance, even outweighing the temperature impact in this case. For the two wafers both annealed at 350 °C, large yield differences were observed due to the CMP differences, with the longer chains suffering more which can be expected. With the optimized CMP, the 10 K and 100 K chains yielded 100% and the 1 million chains yielded > 96%. The yield loss for the other wafer is > 15% for the 100 K chains and > 30% for the 1 M chains. On the other hand, when comparing the red and blue curves that used the same optimized CMP process, one annealed at 350 °C and the other one annealed at 200 °C, the yield loss is not as great. Although lowering the thermal energy supply to the hybrid bonding process did cause ~ 2 % yield drop for the 100 K chains and ~ 5% yield drop for the 1 M chains, the severity is greatly mitigated by meticulous management of the bonding surface topography when adopting the advanced CMP process control.

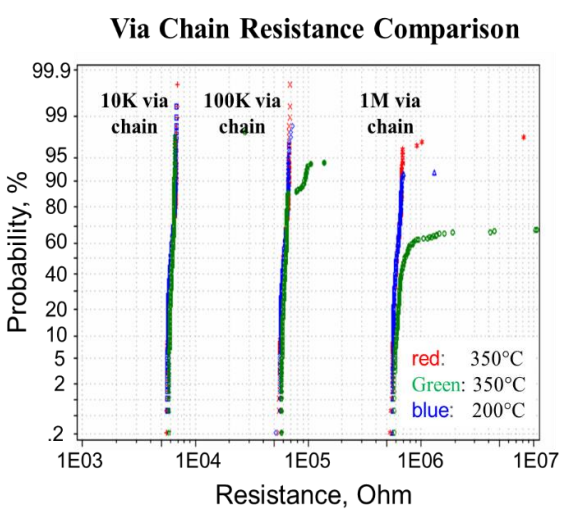

Fig. 5. Via chain resistance distribution comparison for 1 µm pitch daisy chains with different length: 10 K, 100 K, and 1 million links. Three wafers' results were shown: the red curve was for a wafer polished with optimized CMP process and annealed at 350 °C; the green curve was for a wafer polished with fair legacy CMP process and annealed at 350 °C too; the blue curve was for a wafer polished with optimized CMP process and annealed at 200 °C.

C. Benefits of the IF metal diffusion layer in agreesive pitch and temperature scaling

In recent years, several studies have been reported aiming to lower the hybrid bonding temperatures, by means of modifying the Cu grain microstructures to use either nano-twinned Cu with <111> orientation [9] or fine-grained Cu [10] as the bonding layers, or by applying interfacial metal passivation layers on top of Cu to boost Cu diffusivity [9, 11]. For the usage of interfacial capping layers, most of the work published used non-selective PVD deposition on blanket wafers for structural study of Cu interdiffusion. In this article, we used the electroless selective deposition process describe in "EXPERIMENTAL" section D to apply a thin interfacial metal diffusion layer on top of the polished Cu pads. Then finished the electrical test vehicle integration with otherwise identical flow as its no-IF metal counterparts. The benefits of using the interfacial diffusion layer in aggressive pitch scaling and bonding temperature scaling are evaluated below.

Based on the learnings from the above section B, Fig. 5, in order to clearly identify the effects caused by the IF metal, we intentionally chose the non-optimized CMP process for this study. Therefore, the low temperature non-IF reference wafers started with low yields so that we could observe if the IF metal brought in noticeable benefits. Fig. 6 shows whole wafer via chain resistance distribution results for daisy chains with 10,000 and 1 million links at two pitches, 1µm and 0.5 µm. The red curves represent a wafer that used the interfacial metal diffusion layer, and the blue curves stand for another wafer that did not use this interfacial metal. Except for the difference in using the IF metal or not, both wafers were fabricated by the same processes throughout the entire integration flow. After bonding, both wafers were annealed at 280 °C for 2 hours. It shows that, the interfacial metal diffusion layer generally improved the yields of the chains with both moderate and long lengths at both pitches, by approximately 10-15%. Furthermore, for the

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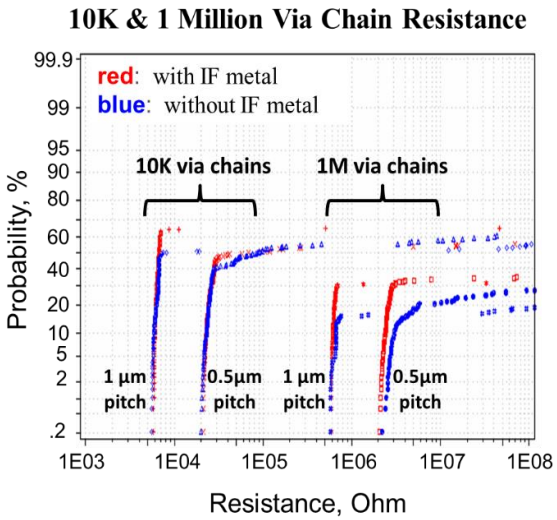

Fig. 6. Via chain resistance distribution for chains with 10,000 and 1 million links. 1µm pitch and 0.5 µm pitch results were both shown for each chain length, for a wafer that used the interfacial metal diffusion layer (red curves), and another wafer that did not use this interfacial metal (blue curves). Except for the IF metal, both wafers went through the same processes throughout the entire integration flow. After bonding, both wafers were annealed at 280°C for 2 hours.

aggressively small 0.5 µm pitch, by using the IF layer, the yield loss when going from the intermediate length of 10K links to the very long 1M links was shrunk from ~ 25% to ~ 15%. This observation shines a light on the possibility of using this interfacial film as one of the enablers for aggressive pitch scaling.

We further examined the impact on yields of this IF metal layer for the very tight 0.5 µm pitch at very long 1 M link length, when the bonding temperature further scaled down to 200 °C. Fig. 7 showed the zoomed-in via chain resistance distributions of 1 million link chains at 0.5 µm pitch, for wafers processed with 280 °C (a) and 200 °C (b) maximal temperature, respectively. For each temperature, we compared wafers bonded with and without using the interfacial layer. Again, leaving out the difference in whether adopting the IF metal or not in the flow, all wafers ran the same processes for the whole integration. At both temperatures, even considering the process variations shown in (b) among wafers with same conditions, the benefits of using the IF metal are still seen in improving the electrical yields for the extremely tight 500 nm pitch and the aggressively long 1M via chain. The yield improvement benefits maintained when the bonding temperature further lowered from 280 °C to 200 °C. It also helped to confirm that the split signal between the two 280 °C bonded wafers shown in Fig. 6 is rather due to the addition of the IF metal than wafer-to-wafer variations.

To our knowledge, this is the first time a work is reported to electrically demonstrate the benefits of using an interfacial metal diffusion layer to improve the yields of million-link length via chains at 500 nm pitch, at low temperatures down to 200 °C. Our

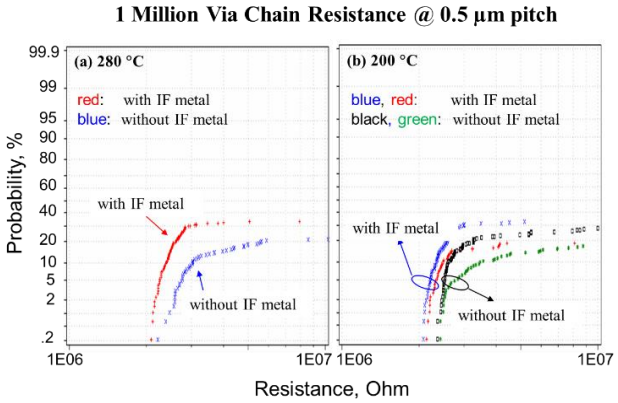

Fig. 7. Via chain resistance distribution comparison for 1 million via link chains at 0.5 µm pitch, for wafers processed with 280 °C and 200 °C temperature limits, respectively, with and without using the interfacial metal layer. (a) 280 °C bonding and post-bonding temperature. (b) 200 °C bonding and post-bonding temperature.

findings showed the possibility of low temperature bonding tight pitch devices in high-volume manufacturing (HVM).

IV. CONCLUSIONS AND NEXT STEP

We used a robust W2W hybrid bonding test vehicle to evaluate two process solutions for enabling hybrid bonding temperature and pitch aggressive scaling. One method is to carefully control the bonding layer topography by using Applied Materials' Reflexion® LK Prime CMP system with advanced process tuning capabilities. Another approach is to enhance the Cu diffusivity at low temperatures by selectively depositing a thin interfacial metal diffusion layer on top of Cu bonding pads with a proprietary electro-less deposition process. By optimizing the Cu/SiCN CMP process with state-of-the-art process control capabilities, a 200 °C bonded wafer only showed slightly dropped yield comparing to its 350 °C reference, demonstrating the paramount importance of CMP process control in ensuring hybrid bonding quality. Intimate Cu bonding at 200 °C was seen by TEM for wafers with or without using the interfacial metal diffusion layer. However, whole wafer electrical yield analysis revealed the benefits of using the interfacial layer to improve yields at low temperatures for aggressive pitches and large-scale bonding chains. To our knowledge, this is the first demonstration of interfacial metal diffusion layer enhancing the electrical yields of 1 million via long chains, which could be a potential technology enabler leading to HVM.

Since the benefits of the IF metal layer were intentionally evaluated by using a low-yield CMP process, and it's shown that CMP control is the most crucial knob for high bonding quality, we are incorporating the interfacial metal layer with the optimized CMP process, aiming to explore the feasibility of combining the two process technology enablers to realize highly yielding low temperature bonding at deeply scaled pitches.

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